Experiment : 6

AIM:- Plot the spectra of ideally sampled signal w.r.t. sampling of Discrete time signals.

Code:

```
clc;
clear;
close all;

t = -0.3:0.0001:0.3;
f1 = 3;
f2 = 23;

fmax = max(f1, f2);

x = cos(f1 * 2 * pi * t) + cos(f2 * 2 * pi * t);

funder = 0.4 * fmax;
fniquist = 2 * fmax;
fover = 7 * fmax;

Tunder = 1 / funder;
Tniquist = 1 / fniquist;
Tover = 1 / fover;

n1 = -0.3:Tunder:0.3;
n2 = -0.3:Tniquist:0.3;
n3 = -0.3:Tover:0.3;

subplot(2, 2, 1);
plot(t, x, 'b', 'LineWidth', 1.2);
hold on;
plot(t, zeros(size(t)), 'k--', 'LineWidth', 1);
hold off;
title('Original Signal');
xlabel('Time (s)');
ylabel('Amplitude');
legend('Signal', 'y = 0', 'Location', 'southwest');
grid on;
```

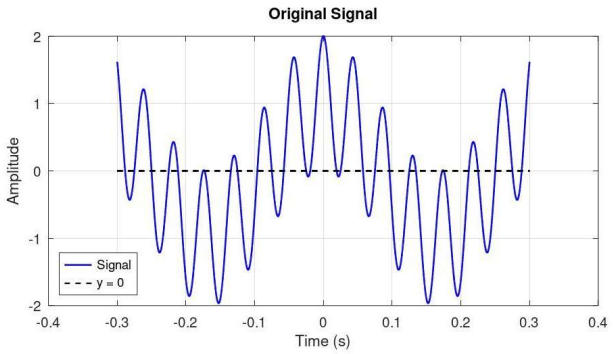
```
xunder = cos(f1 * 2 * pi * n1) + cos(f2 * 2 * pi * n1);
xniquist = cos(f1 * 2 * pi * n2) + cos(f2 * 2 * pi * n2);
xover = cos(f1 * 2 * pi * n3) + cos(f2 * 2 * pi * n3);

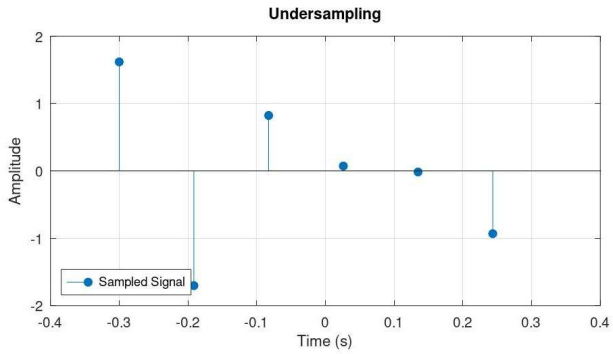
subplot(2, 2, 2);
stem(n1, xunder, 'filled');
title('Undersampling');
xlabel('Time (s)');
ylabel('Amplitude');
legend('Sampled Signal', 'Location', 'southwest');
grid on;

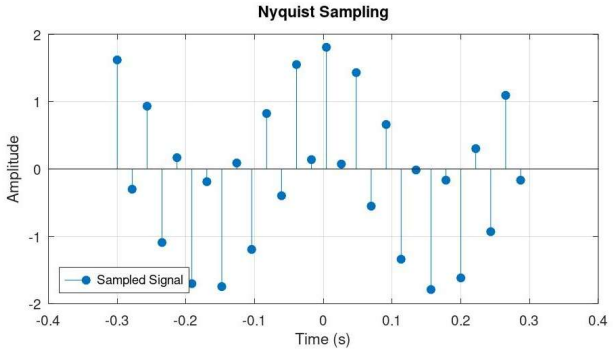
subplot(2, 2, 3);
stem(n2, xniquist, 'filled');
title('Nyquist Sampling');
xlabel('Time (s)');
ylabel('Amplitude');
legend('Sampled Signal', 'Location', 'southwest');
grid on;

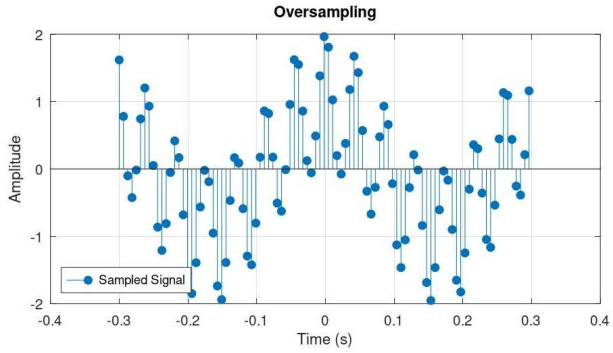
subplot(2, 2, 4);
stem(n3, xover, 'filled');
title('Oversampling');
xlabel('Time (s)');
ylabel('Amplitude');
legend('Sampled Signal', 'Location', 'southwest');
grid on;
```

Output:


Original Signal
Amplitude
Time (s)
Signal
y = 0


Undersampling
Amplitude
Time (s)
Sampled Signal


Nyquist Sampling
Amplitude
Time (s)
Sampled Signal


Oversampling
Amplitude
Time (s)
Sampled Signal